

Lab 3: Pulse on String

Objective

Determine the speed of a pulse on a taut wire and use that to determine the mass per unit length of the wire. Compare that value to an independent measurement of mass per unit length with uncertainty. Agreement between the methods supports the wave speed on a stretched wire equation, $v = \sqrt{T/\rho_L}$. This lab has evolved from a version developed by a former UCA physics student and myself.¹

Equipment List and Setup

Item	Quantity
steel wire	assortment
guitar pickup	1
Oscilloscope	1
Sonometer	1
kilogram masses & hanger	assortment
triple-beam balance	1
meter stick	1
wire cutters	1
stand, 90° connector, & 3-finger clamp	1

Theory

As derived in class, the speed of a wave on a taut string or wire of uniform mass per unit length, is given by,

$$v = \sqrt{T/\rho_L}, \quad (1)$$

where T is the tension in the string in N and ρ_L is the mass per unit length in kg/m. The wire will be plucked with a finger nail or guitar pick. The resulting pulse will propagate down the wire with speed, v , in m/s. The taut wire can be plucked to generate a pulse that will travel down the wire to be detected by the electric guitar pickup coil. The pickup coil consists of tiny magnets (the metal cylinders) wrapped in many many coils of wire. A moving metal wire that disrupts the magnetic field lines will change the flux through the coil. Faraday's Law states that a change in magnetic flux results in an induced voltage in the coil which we can detect with the oscilloscope.

Procedure

We will make use of the potential difference generated when the pulse transverses the region of magnetic field to determine the speed of the pulse on the wire for a given tension. When the

¹T. Odekirk and W.V. Slaton, "Pulse Speed on a Plucked Wire," The Physics Teacher 50, 244-245 (2012).

pulse hits a bridge it is inverted upon reflection and propagates back. Now the pulse is moving in the opposite direction and you will see a corresponding negative voltage generated in the coil. Hence, if you know the distance between the center of the coil and the bridge that caused the reflection you can use an oscilloscope to measure the time between the pulse and its echo. The ratio of the total distance traveled and the total time gives the speed of the pulse!

Our goal is to verify that Equation 1 is correct. If we plot our experimental data as T vs v^2 then the theory says the slope should be, ρ_L , the mass per unit length of the wire. We can experimentally and independently measure the mass per unit length by measuring the mass of a known length of the wire. A comparison of these two independent measures of ρ_L , with uncertainty!, will allow us to determine if Equation 1 is correct.

Experiment Note: The experiment uses steel wires under high tension. **Wear safety glasses!** Be careful not to over-tension the wires and be mindful to stay away from the wire in case it snaps during the experiment. Also, there is a certain finesse required to pluck the wire to get a nice pulse. Nicer pulses = nicer data.

1. There are two gauges of steel wire we will use for this experiment.
2. Your instructor will demonstrate the use of the apparatus. Note that the tensioning arm allows for a convenient doubling of the force applied. **Watch the angle though!**. Pluck the wire in a vertical direction with your finger nail or guitar pick. You should see a jumble of waves on the screen. To capture several pulse-echo events adjust the horizontal and vertical scale as well as the trigger. Turn on the cursors to measure the time difference between the peak and trough of a pulse-echo pair. Get a feel for the system by plucking the wire with various forces. You will want to find a good plucking motion that generates enough motion of the wire to make a measurable voltage difference but not so much force as to change the tension in the wire appreciably!
3. Once you have a good plucking motion take five trials and average the time between the pulse-echo. Use this average time and the distance between the magnet and relevant bridge to compute an average velocity for this tension. Increase the hanging mass by 1kg and repeat the measurement until you have five data points for this wire. Plot the Tension vs the Velocity Squared. Find the slope and intercept with uncertainty for this data. For the 0.73mm diameter wire I got good data using 2-6kg using the tension doubler.
4. Repeat the above data collection for the other steel wire. I didn't take data on the smaller wire with this setup. I suspect you will need smaller tensioning forces.
5. Now use the provided triple beam balance to determine the mass of a known length of wire. With this information compute the mass per unit length of your wire with uncertainty. Does the mass per unit length computed this way agree with the slope of the Tension vs Velocity Squared data? Hint: Good agreement is when the values with uncertainty overlap each other.